

Syu-Huei (Frank) Huang

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EDUCATION

University of Florida (UF)

Aug. 2025 - May 2027 (Expect)

Master of Science in Electrical & Computer Engineering

Gainesville, FL

- GPA: 4.0 / 4.0
- Achievement Award Scholarship
- Selected Coursework:* Applied Machine Learning, Medical Image Analysis, Computational Photography, Autonomous Robotics, Cloud Computer Systems and Applications

National Taiwan University of Science and Technology (NTUST)

Sep. 2021 - Jun. 2023

Master of Science in Electrical Engineering

Taipei, Taiwan

- Selected Coursework:* Color Image Processing, Edge Computing, Design and Application in Internet of Things

Chung Yuan Christian University (CYCU)

Sep. 2017 - Jun. 2021

Bachelor of Science in Electrical Engineering

Taoyuan, Taiwan

- Academic Excellence Award, Outstanding Service Award

PUBLICATIONS

[1] S. -H. Huang, & C. -H. Lu, "Sequence-Aware Learnable Sparse Mask for Frame-Selectable End-to-End Dense Video Captioning for IoT Smart Cameras," *IEEE Internet of Things Journal*, vol. 11, no. 7, pp. 13039 - 13050, 2023.

[2] G. -T. Liu, C. -H. Lu and S. -H. Huang, "Hand-detection with Transferrable Design for Smart Factories," *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Hualien, Taiwan, 2021.

WORK EXPERIENCE

IoT Smart Living Laboratory | Computer Vision, Image/Video Processing, Transformer

Sep. 2021 - June 2023

Graduate Research Assistant

Taipei, Taiwan

- Architected** a sequence-aware learnable mask attention mechanism and custom loss functions to optimize salient feature extraction, surpassing SOTA benchmarks by **12.62%** in BLEU4 and **22.50%** in CIDEr
- Engineered** a lightweight video understanding framework using a novel residual gated network, achieving a **56.90%** reduction in parameters while maintaining competitive inference performance

Intelligent Control Laboratory

Feb. 2020 - May 2021

Undergraduate Research Assistant

Taoyuan, Taiwan

- Autonomous Rescue System:** Developed a real-time fire detection system using OpenCV on Raspberry Pi, achieving **95%** accuracy in vision-based decision-making
- Hardware Integration:** Engineered C++ firmware for Arduino to control actuators with **<1s** latency; awarded Departmental Outstanding Project Award

PROJECTS

VLM-Guided Autonomous Mobile Manipulation System | LLM/VLM, Object Detection, SLAM (ongoing)

Feb. 2026

- Vision-Language Integration:** Developed an end-to-end system using LLMs/VLMs to translate complex natural language commands into actionable robotic tasks, integrating ROS2 for real-time feedback loops
- Autonomous Manipulation:** Implemented SLAM-based navigation and object detection to locate target items in dynamic environments, optimizing robotic arm motion planning for precise pick-and-place execution

Real-time Lane Departure Warning System (Project Leader @ GatorAI)

Sep. 2025

- Led a cross-functional team of 8 to develop an ADAS prototype, establishing a CARLA-based testing pipeline for risk-free validation before deploying a real-time lane detection model on NVIDIA Jetson Nano with CUDA

CT-MR Thorax-abdomen Intra-patient Registration based on Diffeomorphic VoxelMorph

Sep. 2025

- Developed a Diffeomorphic VoxelMorph framework utilizing Stationary Velocity Fields and hybrid loss to handle large deformations, yielding **>100%** MI improvement and reducing folding artifacts to **0.08%** compared to standard baselines

ML-Powered Ship Detection & Feature Reduction | MLP, PCA, Isomap, LLE, t-SNE, PyTorch, Optimization

Sep. 2025

- Developed a ship detection system based on machine learning and dimensionality reduction techniques, achieving up to **95.53%** accuracy by employing Random Forest, MLP, PCA, Isomap, LLE, and t-SNE algorithms

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, MATLAB, Git, Linux, ROS2, AWS EC2, NVIDIA Jetson Nano/TX2, Raspberry Pi, Arduino

Deep Learning & CV: PyTorch, TensorFlow, OpenCV, Scikit-learn, YOLO, Transformers, U-Net, GANs, ResNet